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Absolute (U–Pb) and relative age determinations of intrusive rocks in the Ragunda rapakivi complex, central Sweden

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Abstract

The Ragunda complex (ca 550 km^2) is one of the two major rapakivi occurrences in central Sweden and forms, together with seven smaller complexes, the youngest group (1.53–1.47 Ga) of rapakivi complexes in the Fennoscandian Shield. In contrast to most of the classic Finnish rapakivi plutons the Ragunda complex consists of large amounts of mafic rocks and a substantial volume of syenite and quartz syenite. At the present erosion level, the Ragunda complex includes ca 30% gabbro, 5% syenite and quartz syenite and 65% amphibole–biotite and biotite granites. Minor amounts of hybrid rocks and a large number of acid and basic dykes are found as well. The complex consists of three massifs with distinctive aeromagnetic anomalies. The Western Massif is a centred complex in which magmatic activity began in the west and moved towards the east, forming a linear sequence of approximately circular gabbro, syenite and granite intrusions. The last intrusion, a amphibole–biotite granite, gave a U–Pb zircon age of 1514 ± 5 Ma. The Central Massif is a large centred complex without any obvious linear intrusion sequence, and consists of several consecutively emplaced plutons including a late granite with a U–Pb zircon age of 1505 ± 12 Ma. The Northeastern Massif is a small centred complex consisting of gabbro closely followed by granite. Magma mingling structures are observed throughout the complex, most commonly in contact zones between gabbro and granite. The generation of the Ragunda complex and the small complexes in central Sweden cannot be explained by a lithospheric plate moving over a stationary hot spot. Deep fractures formed in response to an extensional regime probably controlled the emplacement of the complexes. This is supported by the large number of subparallel mafic and acid dykes associated with them. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Aeromagnetic data; Central Sweden; Geochronology; Internal magnetic structure; Magma mingling; Ragunda complex; Rapakivi granite

1. Introduction

Rapakivi granites and related igneous rocks show a worldwide distribution and numerous occurrences have been reported from all continents. They probably represent volumetrically the most important continental silicic intraplate magmatism in the geological history and thus have important bearings on the evolution of the continental crust (Rämö and Haapala, 1995). The overwhelming majority of rapakivi granites is Proterozoic (1.7–1.0 Ga) with a median age of 1.54 Ga and they are almost invariably located in crustal provinces formed in the Palaeoproterozoic.

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The processes that led to the formation of these rocks during a fairly restricted time period are not completely understood.

Rapakivi complexes found in Precambrian shield areas generally occur along roughly linear intracontinental zones of considerable extent. Emplacement of these complexes has been attributed by most authors to Proterozoic continental rifting events, incipient or aborted, (e.g. Anderson, 1983; Emslie, 1985; Windley, 1983; Haapala and Rämö, 1992). It has also been suggested that the formation of some of these rocks may (Pesonen et al., 1989; Hill and Miller, 1990) or may not (Emslie and Hunt, 1990; Haapala and Rämö, 1992) have been related to a continental plate overriding a single hot mantle plume beneath the continental crust.

The distribution of the major rapakivi complexes in the Fennoscandian Shield and neighbouring parts of the East European Platform suggests that they occur in linear belts that meet at a 'triple point' close to the Åland batholith (Fig. 1). The east–west belt extends from the Salmi complex in Russian Karelia through the classic southeastern Finnish rapakivi complexes to the Åland complex in the west. Included in this belt are the plutons in northern Estonia, which are covered by Palaeozoic platform sediments. The southern belt extends from the Riga complex in Latvia (also covered by platform sediments) to the Åland complex in the north. The northern belt extends from the Åland complex northwards to the Nordingrå and Ragunda complexes in central Sweden. Excluding the Salmi complex [1.56–1.54 Ga; Suominen (1991); Neymark et al. (1994)], there appears to be three geographically separated age groups. The oldest of these consists of the complexes in southeastern Finland and northern Estonia, with ages between 1.65 and 1.61 Ga (Vaasjoki, 1977; Vaasjoki et al., 1991, 1993; Suominen, 1991; Kirs et al., 1991; Rämö et al., 1996). To the west of these complexes, there is a roughly N–S trending belt of complexes from the Riga batholith in the south to the Nordingrå complex in the north with ages ranging between 1.59 and 1.56 Ga (Vaasjoki, 1977; Welin and Lundqvist, 1984; Vaasjoki et al., 1988; Idman, 1989; Suominen, 1991; Rämö et al., 1996). Recent age determinations of the Swedish complexes situated to the W–NW of the last mentioned group have revealed that (except the Nordingrå complex) they form the youngest group of rapakivi complexes, ranging between 1.53 and 1.47 Ga (Welin, 1994; Andersson and Neymark, 1994; Andersson, 1997a,b; Claesson and Kresten, 1997; Persson, 1995, this paper).

The classical Fennoscandian rapakivi granite complexes and their associated basic rocks have been in the focus of interest since the final decades of the nineteenth century (e.g. Sederholm, 1891; Högbom, 1893; Lundbohm, 1893; Wahl, 1925; Sahama, 1945). In contrast to the Finnish rapakivi complexes (e.g. Vorma, 1976; Haapala, 1977; Vaasjoki, 1977; Nurmi and Haapala, 1986; Haapala and Rämö, 1990; Rämö, 1991), the Swedish complexes have only been studied sporadically and little modern work has been reported until recently (e.g. Kornfält, 1976; Welin and Lundqvist, 1984; Andersson and Neymark, 1994; Persson, 1995, 1996; Lindh and Johansson, 1996; Andersson, 1997a,b). The Ragunda complex is one of the two large Swedish rapakivi complexes situated in central Sweden (Figs. 1 and 2). The rocks of the complex are well exposed in the steep banks of river valleys, which are locally inaccessible due to rough topography. As early as 1847, the rocks of the Ragunda complex were reported by Erdmann (1847). The Ragunda complex was noted by Lundbohm (1893) and in the description of the geological map of Västernorrland county [Lundbohm (1899); scale 1:500 000]. The geology of the Ragunda complex was described and discussed in more detail by Högbom (1893, 1894, 1899, 1909). Högbom (1899, 1909) argued that the Ragunda complex was a laccolith. Kornfält (1969, 1976) published a detailed map of the complex and showed that the laccolith hypothesis was incorrect. Kornfält (1976) reported a Rb–Sr age of 1320 ± 30 Ma for the syenites and granites of the complex. The Rb–Sr age has, however, been questioned as the same method used in dating of the Nordingrå rapakivi complex resulted in too young an intrusion age compared to the U–Pb dating (Welin and Lundqvist, 1984).

The chronological aim of this study was to determine whether the younging of the

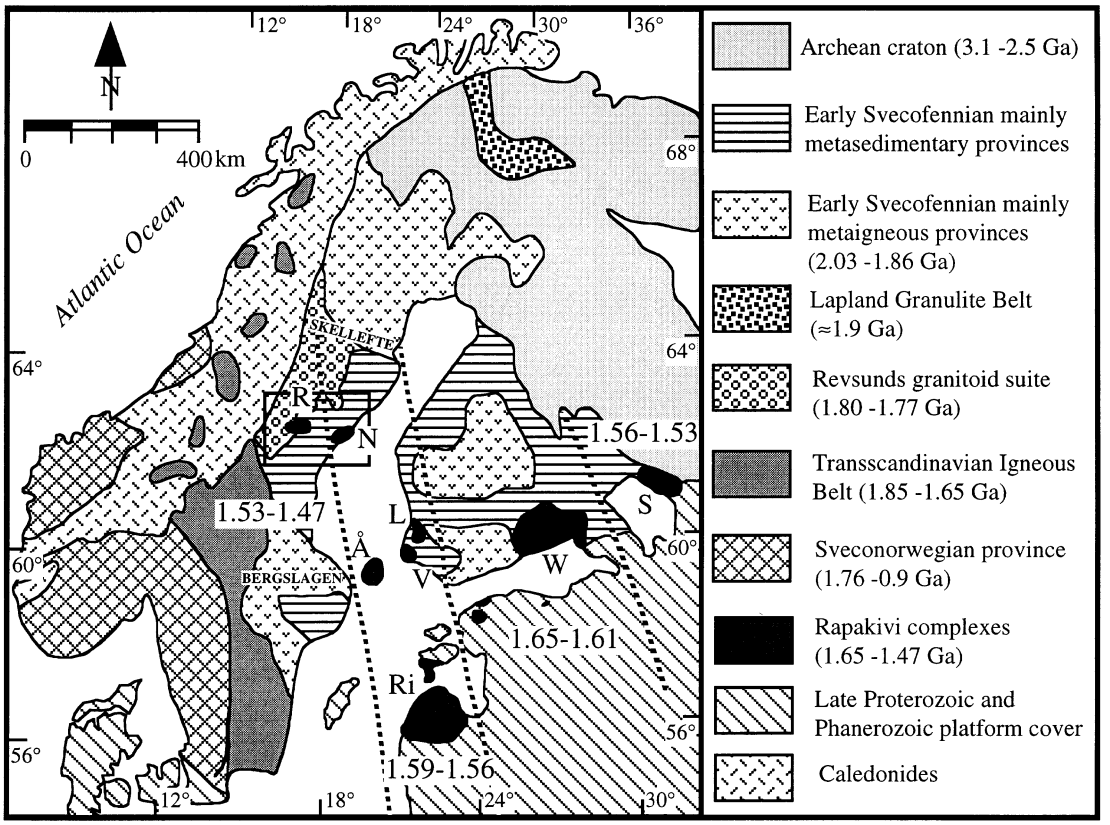


Fig. 1. Tectonic map showing the major rapakivi complexes of the Fennoscandian Shield and neighbouring parts of the East European Platform. The thick vertical broken lines indicate the age (in Ga) distribution of rapakivi intrusives. The area within the frame is shown on a larger scale in Fig. 2. Å, Åland; L, Laitila; N, Nordingrå; R, Ragunda; Ri, Riga; S, Salmi; V, Vehmaa; W, Wiborg. Note that the five southernmost rapakivi occurrences are covered under the Phanerozoic platform cover.

Fennoscandian rapakivi complexes continues towards the west. Simultaneously with this work, dating efforts of the smaller complexes were initiated (Andersson and Neymark, 1994; Andersson, 1997b) and these datings have suggested the existence of a young group of rapakivi intrusions. The internal age relations (intrusion sequence) in the Ragunda complex was another target for the present research. This part of the work is based on new mapping, interpretation of aeromagnetic data and magnetic susceptibility measurements.

2. Geological setting

The bedrock of central Sweden and south-central Finland is dominated by the rocks of the Bothnian Basin. The area is situated between two volcano-sedimentary districts, the Skellefte district in the north and the Bergslagen district in the south, and bounded in the west by the Caledonides and the Transscandinavian Igneous Belt (Fig. 1). The Swedish part of the Bothnian Basin is dominated by early-Svecofennian metasediments, mainly marine metagreywackes (Lundqvist et al., 1990), which are metamorphosed to varying degrees. In large areas, extensive migmatization has occurred. Four distinct generations of granitoids have been identified in the basin (cf Fig. 2). These are traditionally described as early-, late-, post- and anorogenic relative to the ca 1.9–1.8 Ga Svecofennian Orogeny.

Early-Svecofennian granitoids [1.94–1.84 Ga; Delin (1993); Wasström (1993); Welin et al.

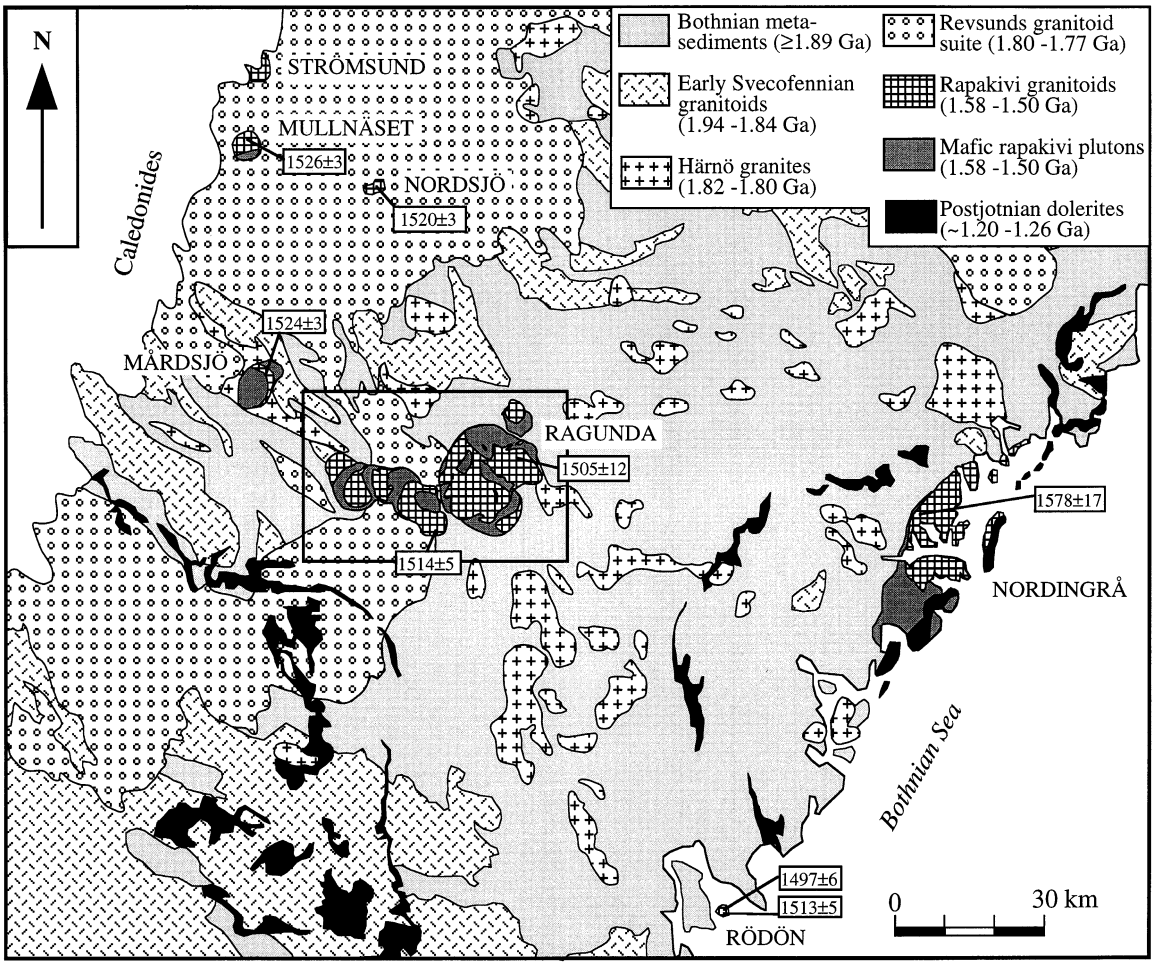


Fig. 2. Simplified geological map of central Sweden. Modified after Lundegårdh et al. (1984) and Lundqvist et al. (1990). The area within the frame is shown on a larger scale in Fig. 3. Ages after Welin and Lundqvist (1984), Nordingrå; Andersson and Neymark (1994), Mullnäset, Mårdsjö; Welin (1994), Rödön; Persson (1995), this paper, Ragunda; Andersson (1997b), Rödön, Nordsjö.

(1993); Claesson and Lundqvist (1995)], comprise a calc-alkaline suite of mainly tonalite and granodiorite. Volcanic rocks are rare and mostly mafic. The late-Svecofennian granitoids [1.82–1.80 Ga; Claesson and Lundqvist (1995)] were intruded after the main phase of Svecofennian deformation and regional metamorphism. They are predominately granitic, peraluminous, associated with abundant pegmatites and show S-type characteristics. The post-orogenic, 1.80–1.77 Ga, Revsund granitoid suite (Claesson and Lundqvist, 1995) occupies the area towards the west. This suite shows a wide range in composition, from granite to quartz monzodiorite (Persson, 1978; Lundqvist et al., 1990), and is mostly I-type, with local A- and S-type granite signatures (Claesson and Lundqvist, 1995; Persson, 1978).

The western Fennoscandian rapakivi suite (cf Fig. 2), consists of two major (Ragunda and Nordingrå) and at least seven minor complexes (Strömsund, Mullnäset, Nordsjö, Mårdsjö Rödön and, outside area of Fig. 2, Noran and Strömsbro). As is typical for most of the classic Finnish rapakivi complexes, silicic rocks are the predominant rock types. However, in the Swedish complexes, there is also a significant amount of mafic rocks.

The Ragunda complex is exceptional in containing a significant volume of syenite and quartz syenite as well.

The rocks of the rapakivi group are also termed Subjotnian (Högbom, 1909), as they often form the basement of the Jotnian sedimentary successions of the shield. These were deposited after the emplacement of the rapakivi complexes and before the intrusion of the Postjotnian dolerite dykes, at ca 1.50–1.20 Ga. Continental Jotnian sedimentary rocks are abundantly associated with the Laitila and Salmi rapakivi granites. Jotnian sandstone has also been reported from the Nordingrå area (e.g. Sobral, 1913; Lundqvist et al., 1990). In the Ragunda complex, sandstone has been found at one locality, in the western part (Western Massif — see below) filling a >4 m wide fissure.

A large number of Postjotnian dolerite dykes occurs in central Sweden (Fig. 2). These dykes have K–Ar dates of 1200–1250 Ma [Welin and Lundqvist (1975); Welin (1979); cf also Suominen (1991)]. A flat lying dolerite dyke intrudes the syenitic rocks of the Ragunda complex. These dolerites have been referred to as the Central Scandinavian Dolerite Group (Gorbatschev et al., 1979).

3. The Ragunda complex

The Ragunda complex covers an area of ca 550 km^2 (Fig. 3). It was previously mapped as a single complex with a small satellite, 'Holmstrand complex', situated northeast of the main Ragunda complex (Kornfält, 1976). The present study includes sampling for isotopic analyses, new mapping aided by new road cuts and the new topographic and economic maps of Sweden at the scale of 1:50 000 and 1:10 000, respectively. A revised map of the Ragunda Complex (Fig. 3) was compiled from Kornfält's observations, the new field observations and aeromagnetic data. The Ragunda complex consists of three distinct units: the Western; Central; and Northeastern Massifs.

The Ragunda complex cuts sharply across the surrounding Svecofennian plutonic and supracrustal rocks. The Central Massif and the Northeastern Massif are surrounded by well-preserved metagreywackes. To the west, the Central Massif abuts on migmatized metagreywackes. A small body of the late-Svecofennian Härnö granite is located at the eastern border of the Central Massif. The Western Massif intrudes the Revsund granite [Figs. 3 and 4(a)], and the westernmost part intrudes into migmatized metagreywacke and a small body of an early-Svecofennian granitoid. The contacts between the complex and the country rocks seem to be approximately vertical [Fig. 4(b)]. Brecciation of the older country rocks is common and xenoliths of these rocks frequently occur in the contact areas.

The Ragunda complex consists of three major rock groups:

(1) 30% gabbro (including a small amount of leucogabbro);
(2) 5% syenite and quartz syenite; and
(3) 65% amphibole–biotite and biotite granites.

Some minor amounts of hybrid rocks are found in addition to numerous acid and basic dykes.

3.1. Gabbro

The gabbro is a fine-grained or fine- to medium-grained, blackish grey to dark grey rock. Locally, fine- to medium-grained gabbro occurs containing white megacrysts (2–12 mm) of plagioclase. Zoned plagioclase is the predominant mineral in the gabbro; the core is composed of labradorite and the marginal zone of andesine or oligoclase (Kornfält, 1976). The mafic minerals are mainly clinopyroxene (ferroaugite), orthopyroxene (ferrohypersthene) and biotite (Kornfält, 1976). The pyroxenes are locally altered to amphibole. Approximately 65% of the analysed samples have normative olivine, although olivine was found only sporadically in thin section. Ilmenite, magnetite and pyrite are always present. Intrusion of granite into the gabbro is common. Magma mingling structures (pillows of gabbro net-veined by granite, Fig. 5) have been observed; hybrid rocks have also been found. In the Northeastern Massif, a small isolated outcrop of leucogabbro occurs. Compared to the gabbro, the leucogabbro is more coarse-grained (plagioclase grains are typically >1 cm in length) and has more plagioclase (ca 65–75 vol%). In the northeastern part of the Central Massif, a

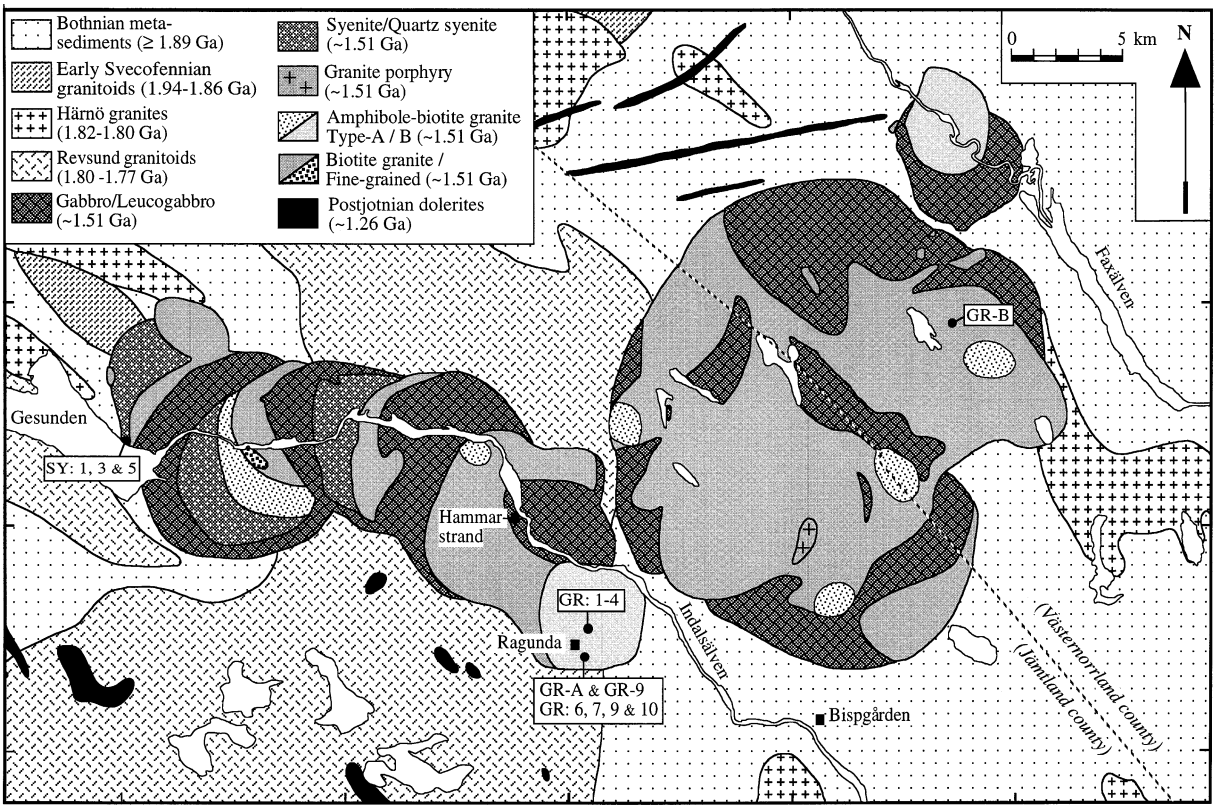


Fig. 3. Geological map of the Ragunda complex. Compiled after Kornfält (1969, 1976), new field observations and aeromagnetic data (Fig. 6). Sample localities for U–Pb and Rb–Sr isotopic analyses are indicated by numbers in rectangles.

gabbro outcrop consists of patches of the coarse-grained leucogabbro within the normal gabbro. There are no sharp contacts between these two rock types.

3.2. Syenite and quartz syenite

The syenitic and quartz syenitic rocks occur only in the Western Massif and are found as three separate intrusions. They are typically medium- to coarse-grained and greyish green to greyish red. The syenite and quartz syenite consist mainly of perthitic orthoclase (Kornfält, 1969); film, vein and patch perthite occurs. Locally the plagioclase (oligoclase) patches merge into larger crystals that show the same optical orientation as the minor patches within one single potash feldspar crystal. In the quartz syenite, small amounts of interstitial plagioclase occur. Quartz is present in all analysed thin sections. The mafic minerals are mainly clinopyroxene (ferroaugite), amphibole and fayalite ($Fa_{93.5}$) (Kornfält, 1976). The opaque minerals are mainly ilmenite and pyrrhotite together with small amounts of chalcopyrite and pyrite. Titanomagnetite occurs sporadically. A few percents of biotite is also present. Accessory minerals are apatite, which locally exceeds 1 vol%, fluorite and occasionally zircon.

A contact between syenite and biotite granite has been found in the westernmost part of the Western Massif. It is sharp and dips ca 45° to the west. Close to the contact, syenite and gabbro xenoliths are found in the granite.

3.3. Amphibole–biotite granite

Two major types of amphibole-bearing granite occur in the Ragunda complex, as already noted

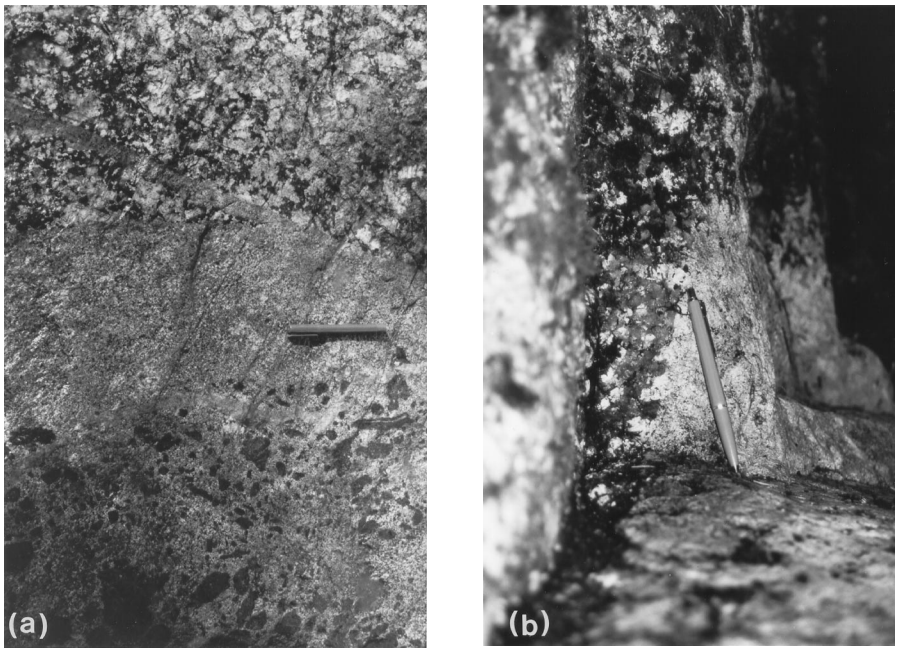


Fig. 4. (a) Contact relations between the Ragunda granite (below) and the Revsund granite (above). The photo is taken from a horizontal outcrop, ca 1 km south of the village Ragunda in the Western massif. The contact is sharp, with the Ragunda granite (amphibole–biotite granite) cutting the potassium feldspar megacrysts of the Revsund granite. There is also a very narrow (2 cm) apophyse from the Ragunda granite cutting the Revsund granite. At this locality, the amphibole–biotite granite is fine-grained and pale-red in a 20 cm wide contact zone. There is a zone of fine-grained, light-red amphibole–biotite granite crowded with small (0.5–20 cm) inclusions of gabbro below the pencil. (b) Vertical contact between Ragunda biotite granite (right) and Revsund granite (left), ca 1.5 km south-west of the village of Ragunda. The biotite granite shows a fine-grained contact zone against the Revsund granite.

by Kornfält (1976). Their distinction is here based on both modal and chemical composition of the rocks (A.I. Persson, unpublished data). The first type (Type-A) always contains either olivine or secondary minerals after olivine and has both a monoclinic and a triclinic potassium feldspar (Kornfält, 1969). In this way, it resembles the syenitic rocks. Alkali amphibole (riebeckite) occurs as discrete small grains but mostly as overgrowths on hornblende. Vein and patch perthite are both present and the rock shows a hypersolvus texture.

The second type (Type B) of the amphibole-bearing granite is subsolvus like the biotite granite (see below). Plagioclase (oligoclase) occurs mainly as discrete crystals, olivine or secondary minerals after olivine are not present and the quartz content is typically low compared to that of Type A. The Type-B amphibole-bearing granite was described only from the Northeastern Massif (Holmstrand) by Kornfält (1976). For this work, a large Type-B amphibole–biotite granite intrusion was found in the eastern part of the Western Massif, around the village of Ragunda; a comparable but smaller intrusion is found in the E–NE part of the Central Massif (Fig. 3). The amphibole–biotite granite in Ragunda consists of nearly equal amounts of hornblende and biotite (some of the Type-A amphibole granites also contain nearly equal amounts of them). Allanite is a typical accessory mineral in Type-B.

3.4. Biotite granite

The most abundant granitic rock is a medium- to coarse-grained biotite granite, which varies in colour, grain size and texture. It is typically porphyritic with abundant rectangular potassium feldspar megacrysts (up to 1 cm) in a medium-grained

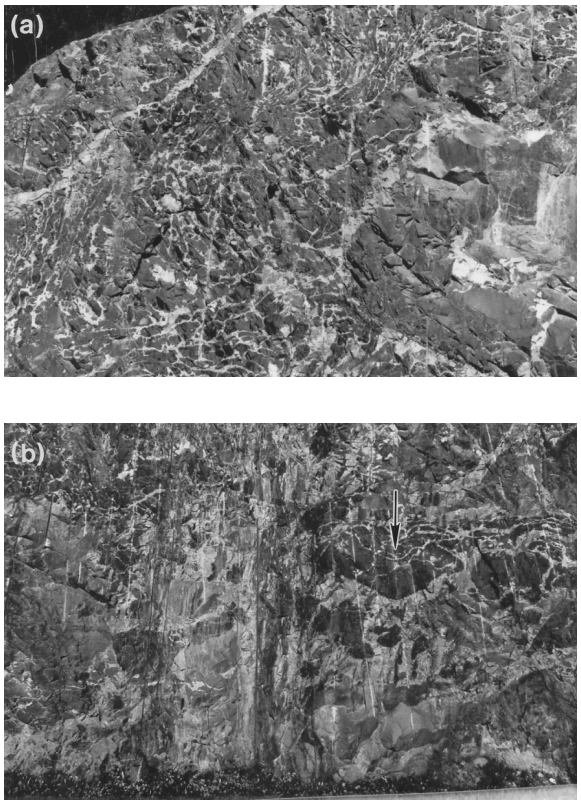


Fig. 5. Magma mingling structures in the Ragunda rapakivi complex. Photo from a road cut at 'Prästberget', ca 1 km south–southwest of the village Hammarstrand. (a) Gabbro net-veined by granite. The large gabbro pillow in the left part measures 7 m in height. (b) Photo taken to the left of the large gabbro pillow showing more angular gabbro and hybrid (light grey) fragments in granite. The arrow points to a gabbro enclave (ca 4.5 m across) which is proposed to have a downward sinking structure (smooth curved lower part and a rough upper part penetrated by granite).

matrix. Granophyric, equigranular and fine-grained varieties also occur. The megacrysts are locally mantled by plagioclase. Plagioclase in the biotite granite occurs as exsolved patches in the potassium feldspar; however, the majority of the modal plagioclase is found as discrete crystals. In all granite varieties (including the amphibole–biotite granites), quartz commonly appears as 'drop' quartz with crystals up to 8 mm in diameter. Micropegmatitic intergrowths of quartz and potassium feldspar are common, typically in the margins of the potassium feldspar crystals. In the most porphyritic varieties, large (ca 1 cm) subhedral or sometimes euhedral potassium feldspar crystals occur in a medium-grained (1–2 mm) granophyric matrix. The quartz occurs as large (2–8 mm), approximately euhedral grains ('drop' quartz) and small (<1 mm) anhedral grains intergrown with the potash feldspar, forming a granophyric texture (Kornfält, 1976). Anhedral quartz grains also occur between the potassium feldspar crystals. Magnetite and ilmenite are found in all samples. Zircon and fluorite are accessory minerals in all samples, allanite is less common. Fluorite amounts to 1 vol%.

A fine-grained, red, porphyritic granite has been found in the Western Massif within the Ragunda Complex (cf Fig. 3). This fine-grained biotite granite is generally pale red, with sparse euhedral phenocrysts (1–3 mm) of potassium feldspar. Euhedral phenocrysts (1–2 mm) of quartz are more abundant. Plagioclase occurs locally as sericitized phenocrysts. The matrix is fine-grained with micropegmatitic intergrowths of quartz and potassium feldspar. The phenocrysts comprise 5–15% of the volume of the rock.

Aplitic granite occurs with biotite granite in the gabbro, forming narrow dykes. Small narrow pegmatite dykes occur rarely in the gabbro breccia. Only a few aplite dykes have been observed in the granites. The aplites are narrow (<1 m), greyish red to red, and almost devoid of mafic minerals. No pegmatite dykes have been found in the complex.

A granite porphyry has been mapped in the Central Massif (Fig. 3). It is fine-grained, greyish brown to brownish grey, with ca 1 cm long, pink, euhedral phenocrysts of potassium feldspar. Phenocrysts of light green, euhedral plagioclase crystals ca 5 mm in length and drop quartz measuring 3–5 mm in diameter also occur, as well as biotite and hornblende phenocrysts.

3.5. Acid and basic dykes

The Ragunda complex and surrounding country rocks are cut by basic and acid dykes. These dykes can be considered as part of the Ragunda rapakivi complex. During field work for the present contribution >40 new dykes were found. The two largest dykes (12 m and 8 m wide) found within the

complex are composite. The majority of the acid dykes are quartz–feldspar porphyries. Dyke widths range from a few decimetres to 12 m, and are typically ca 2 m. The majority of the dykes are found in the Western Massif, probably due to the better degree of exposure of this part of the Ragunda complex. In the Western Massif, the dykes generally trend between E–W and NW–SE and dip vertically or steeply to the N–NE.

The dykes show an overall, weakly defined E–W trend. The largest dyke of the Ragunda complex is found in the Central Massif and strikes E–W. It contains 2 m wide, basic, bluish grey margins and a 8 m wide, central brownish red quartz–feldspar porphyry. The second largest dyke (8 m wide) is found in the Western Massif. This dyke is composed of 0.5 m wide grey, fine-grained plagioclase porphyritic margins and a 7 m wide, central quartz–feldspar porphyry. At the silicic–basic contacts, there are zones of mingled rocks.

Most of the dykes in the complex have sharp, chilled contacts against the rapakivi country rocks. However, in the westernmost gabbro intrusion of the Western Massif, a few narrow (0.3–2.0 m wide) quartz–feldspar porphyries show sharp and sinuous contacts with the gabbro. The sinuous contact points to a short time interval between the emplacement of the gabbro and the intrusion of the quartz–feldspar porphyries.

4. Aeromagnetic investigation

Magnetic susceptibility measurements have been performed on the different rock types of the Ragunda complex and the country rocks, mainly for rock type correlation, but also to elaborate on the hypothesis whether the rocks present at the surface are responsible for the aeromagnetic anomaly pattern. A small portable, battery-operated, susceptibility meter 'JH-8' was used for measurements on the outcrops. The susceptibility meter JH-8 is an 'active instrument' which produces a magnetic field and then measures the response of the ground to the applied field. The sensing depth of this instrument is fairly shallow, a few centimetres. Measurements were performed at 122 localities. From each locality (most of them road cuts), 10–15 measurements were made and then averaged.

The susceptibility measurements on the gabbro gave high values, typically between 0.02 and 0.05 SI units. The syenites show low to moderate (0.001–0.02 SI), the granites generally low (0.001–0.008 SI) and the country rocks very low values. The metagreywackes range between 0 and 0.0005 SI units and the Revsund granite between 0 and 0.0003 SI units. The gabbro of the Western Massif is generally higher (0.04 SI) than the gabbro of the Central Massif (0.02 SI) and the Northeastern Massif (<0.015 SI). In the Western Massif, values as low as 0.002–0.02 SI units were obtained for the fine- to medium-grained, plagioclase–porphyritic gabbro that appears to be more common in the Central Massif than in the Western Massif. In the Northeastern Massif, very low values were obtained from the leucogabbro (0.0005 SI) and from a fine-grained gabbro variety containing abundant plagioclase phenocrysts (0.0004 SI).

The granites generally have susceptibilities around 0.0035 SI units, but at some localities, very low values (0.0001 SI) were measured. The lowest values were obtained for the amphibole-bearing granite in the Northeastern Massif and the westernmost granite intrusion of the Western Massif. High values were measured for the amphibole-bearing granite south and southwest of the village of Ragunda (0.008 SI). The susceptibility is very low (0.0002–0.001 SI) in the centre of this amphibole-bearing granite (NE of Ragunda village). The westernmost syenitic intrusion also yields very low values (0.0005 SI). The second syenitic intrusion from the west yielded a large range in susceptibility: in the northern part of the body, values between 0.002 and 0.03 SI units were measured, and in the southern part of it, values as low as 0.0005 SI units were obtained. The third syenitic intrusion showed values between 0.003 and 0.02 SI units. There is thus a clear correlation between rock type and magnetic susceptibility.

The aeromagnetic data quoted here were acquired by LKAB Prospekterings AB in the late 1970s. The flight altitude was 30 m and the direction was E–W. For this study, the magnetic patterns were interpreted from aeromagnetic maps at

the scale 1:200 000 with a pixel-size of 50 × 50 m. A digital outprint of the data was produced by the Geological Survey of Sweden with a pixel-size of 200 × 200 m (Fig. 6).

The aeromagnetic map reveals important information on the shape and relative age of the various magnetic structures (truncation relations). The Ragunda complex clearly stands out as a magnetic anomaly (the brighter the pattern, the higher the intensity of the magnetization; Fig. 6) and the three massifs can be easily separated. Overall, the country rocks have a very low magnetization level. The intensity variation in the aeromagnetic map correlates well with the outcrop magnetic susceptibility measurements. The Western Massif consists of a grossly linear intrusion sequence composed of approximately circular gabbro, syenite and granite intrusions.

The magnetic pattern of the Central Massif forms a 'cloudy' structure on the magnetic map (Fig. 6) and does not show any obvious linear intrusion sequence. The highest magnetization levels are found mostly near the margins that consist of gabbro. Gabbro also occurs in a NW-oriented band across the central part of the Central Massif and shows up as an area of increased magnetization. In the low magnetic areas, several approximately circular intrusions can be distinguished.

The Northeastern Massif is clearly distinguished as a separate magnetic anomaly northeast of the Central Massif. It is not possible, however, to distinguish between the two major rock types, gabbro and granite. This is in line with the magnetic susceptibility measurements on the gabbro and granite.

5. Isotopic investigation

Two granite varieties of the Ragunda complex were sampled for U–Pb zircon work and one

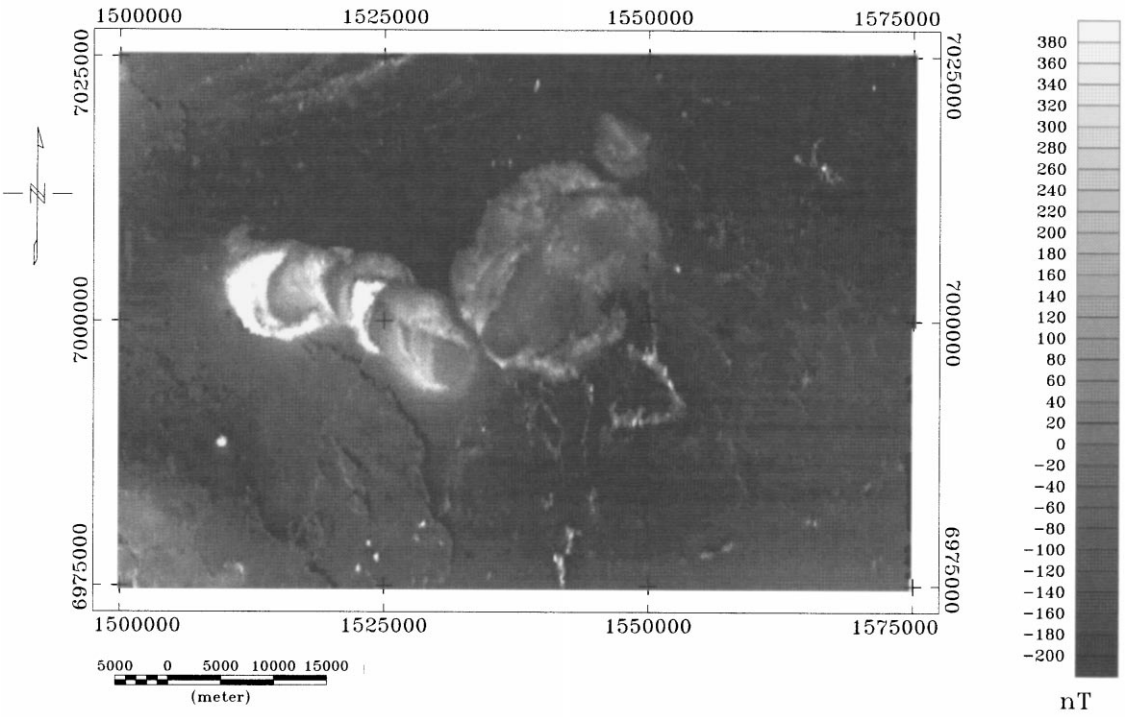


Fig. 6. Aeromagnetic map of the total field of the Ragunda area. Co-ordinates in the Swedish National Grid. Published with the permission of the Geological Survey of Sweden (No. 00-1238/96).

granite and one syenite for Rb–Sr whole-rock work. The sample localities are shown in Fig. 3.

5.1. U–Pb zircon work

5.1.1. Sample material

Two granite intrusions were sampled for U–Pb zircon age determination. Sample GR-A is a red, coarse grained amphibole–biotite granite from the Western Massif (Fig. 3). It is porphyritic with rectangular potassium feldspar megacrysts (typically up to 1 cm) in a medium-grained matrix. The zircons from sample GR-A consist of euhedral crystals with sharp, well-developed tetragonal prisms and simple pyramidal endings. Generally, most crystals have a length-width ratio of 3:1, but long-prismatic crystals with a ratio 6:1, and short-prismatic crystals with a ratio 2:1 also occur. The majority of the zircons are transparent and pale yellowish brown. They do not contain any observable older cores or later overgrowths. Three fractions from the smallest size fraction (70–50 μm) were selected for analysis; they were nearly colourless and transparent. One of the fractions (GR-A; 70–50 I) contained long prismatic crystals. A previous sample, GR-9, was taken from the same locality as GR-A. The analysed zircons from this sample were selected from three size fractions.

The second sample GR-B is a light-red to red, medium grained, equigranular biotite granite from the Central Massif. The zircons from sample GR-B are similar to those in GR-A with sharp, well developed euhedral prisms and simple pyramidal endings. In contrast to GR-A, the zircons from GR-B generally have a length–width ratio of 2:1. The majority of the zircons is similar to those in GR-A-transparent and pale yellowish brown. They do not contain any observable older cores or later overgrowths. Six fractions from three size fractions were selected for analysis. They were nearly colourless and transparent.

5.1.2. Analytical methods

The U–Pb analytical work for samples GR-A and GR-B was performed at the Laboratory for Isotope Geology, Swedish Museum of Natural History, Stockholm. Zircons from sample GR-9 were analysed at the Institut für Geologie und Dynamik der Lithosphäre, University of Göttingen, Germany. The mineral separations were carried out at the Department of Geology, Lund University, Sweden.

The zircon fractions were separated from the crushed and ground rocks using a continuous sieve, a Wilfley table, heavy liquids and a Frantz isodynamic magnetic separator. The zircons were removed stepwise and acting non-magnetically at a current of 1.7 A, and 2.5° vertical and 5° horizontal tilt. The concentrates were further split into size-fractions from which the most transparent, prismatic and inclusion-free crystals were hand-picked. All the hand-picked fractions from Stockholm, with one exception (GR-A: 50–70 I), were air abraded (cf Krogh, 1982) for a few hours and the best grains were picked out for chemical preparation. The three fractions from GR-9 were not air abraded. At both laboratories the chemical procedures, including the decomposition of the crystals and the extraction of uranium and lead, were performed essentially according to the method described by Krogh (1973). In Göttingen, a mixed ^{208}Pb–^{235}U spike was used and the measurements were made on a Teledyne NBS-type 12″90° solid source mass spectrometer [additional details are given by Persson et al. (1983)]. In Stockholm, a ^{233}–^{235}U tracer was added to the solution, which then was aliquoted. One aliquot was spiked with a ^{208}Pb tracer. Both uranium and lead were loaded on Re filaments using nitric acid- and phosphoric acid/silica gel techniques, respectively. Both uranium and lead isotopic ratios were measured on a Finnigan-MAT 261 mass spectrometer with five Faradays cups. Mass-fractionation correction of 0.10% amu^{-1} for Pb was made, based on measurements of NBS SRM 981. The ratios of Todt et al. (1993) were used. The mass fractionation for U was measured by means of the ^{233}–^{235}U spike. Common lead in the sample was corrected for using the growth curve of Stacey and Kramers (1975) with the following ratios: $^{206}Pb/^{204}Pb = 16.1$, $^{207}Pb/^{204}Pb = 15.4$ and $^{208}Pb/^{204}Pb = 35.8$. The calculations of elemental and isotopic compositions were calculated with PBDAT (Ludwig, 1980, 1991a) and concordia intercept ages were calculated with ISOPLOT (Ludwig, 1991b). The decay constants recom-

mended by the IUGS Subcommission on Geochronology (Steiger and Jäger, 1977) were applied. The total lead contamination was, on average, 13–20 pg for GR-A and GR-B. For the analyses in Göttingen it was 1.8–2.2 ng.

5.1.3. Results

The analytical results from the U–Pb analyses are summarised in Table 1 and displayed on concordia plots in Fig. 7. Three zircon fractions from sample GR-A yield a discordia line with an upper intercept age of 1513 ± 9 Ma, a lower intercept age of 200 ± 207 Ma and an MSWD (mean square of weighted deviates) of 0.31 [Fig. 7(a)]. The upper intercept is interpreted as the intrusion age of the granite. This interpretation is supported by the almost concordant fraction 3. The lower intercept cannot be given any geological interpretation. Six zircon fractions from sample GR-B yield a discordia line with an upper intercept age of 1505 ± 12 Ma, a lower intercept age of 300 ± 211 Ma and an MSWD = 0.76 [Fig. 7(b)]. This upper intercept age is interpreted to date the intrusion.

The three zircon fractions from sample GR-9 yield a discordia line with an upper intercept age of 1460 ± 82 Ma, a lower intercept age of 82 ± 251 Ma and a MSWD = 1.1. This is not a well-defined discordia as the data are >35% discordant and grouped closely together on the discordia. When plotting the data from the fractions of GR-9 together with the data from GR-A, all fractions fall on a well-defined discordia line with an upper intercept age of 1514 ± 5 Ma, a lower intercept age of 234 ± 25 Ma and a MSWD = 0.85 [Fig. 7(c)]. This age is identical to that obtained from only GR-A. The error is reduced and therefore I will use this composite age and error as the

Table 1
Results of U–Pb zircon analyses of the Ragunda complex, central Sweden

Zircon fraction[a] (μm)	Fraction weight (mg)	Measured ratios		Concentrations		Calculated ratios[d]			Model ages (Ma)			Rho[e]
		$^{208}Pb/^{206}Pb$[b]	$^{206}Pb/^{204}Pb$[c]	U (ppm)	Pb rad (ppm)	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	
Even-grained biotite granite, GR-B (700910/154710)[f]												
1. 50–70 I ab	0.019	0.2524	2329	1321	378	0.2486 ± 5	3.203 ± 22	0.09342	1432	1458	1496 ± 12	0.40
2. 50–70 II ab	0.037	0.2488	6562	1482	420	0.2460 ± 2	3.161 ± 5	0.09319	1418	1448	1492 ± 2	0.70
3. 70–100 I ab	0.024	0.2381	3138	1257	344	0.2403 ± 8	3.086 ± 17	0.09316	1388	1429	1491 ± 8	0.61
4. 70–100 II ab	0.032	0.2395	5304	1030	293	0.2488 ± 8	3.205 ± 18	0.09343	1432	1458	1497 ± 8	0.61
5. 70–100 III ab	0.017	0.2595	872	701	192	0.2409 ± 9	3.079 ± 18	0.09272	1391	1427	1482 ± 9	0.63
6. 100–150 ab	0.037	0.2287	4255	1097	318	0.2561 ± 3	3.302 ± 15	0.09353	1470	1482	1499 ± 8	0.36
Hornblende–biotite granite, GR-A (699425/153075)[f]												
1. 50–70 I	0.023	0.2524	1288	642	173	0.2361 ± 5	3.036 ± 38	0.09326	1366	1417	1493 ± 22	0.44
2. 50–70 II ab	0.063	0.2244	9971	592	165	0.2466 ± 3	3.191 ± 5	0.09384	1421	1455	1505 ± 2	0.65
3. 50–70 III ab	0.009	0.1878	2172	256	73	0.2618 ± 14	3.399 ± 22	0.09416	1499	1504	1511 ± 7	0.81
Hornblende–biotite granite, GR-9 (699425/153075)[f]												
1. 74–106	3.880	0.2763	658	814	153	0.1657 ± 6	2.062 ± 15	0.09024	988	1136	1431 ± 4	0.52
2. 106–150	5.627	0.2847	691	803	160	0.1743 ± 7	2.180 ± 16	0.09072	1036	1175	1441 ± 4	0.54
3. >150	7.000	0.3558	381	700	148	0.1805 ± 6	2.248 ± 24	0.09034	1070	1196	1433 ± 4	0.46

[a]ab, abraded fraction.
[b]Corrected for blank, mass fractionation.
[c]Corrected for mass fractionation.
[d]Corrected for blank, mass fractionation, and common lead; errors refer to the last significant digits.
[e]$^{207}Pb/^{235}U$–$^{206}Pb/^{238}U$ error correlation.
[f]Co-ordinates given relative to the Swedish National Grid.

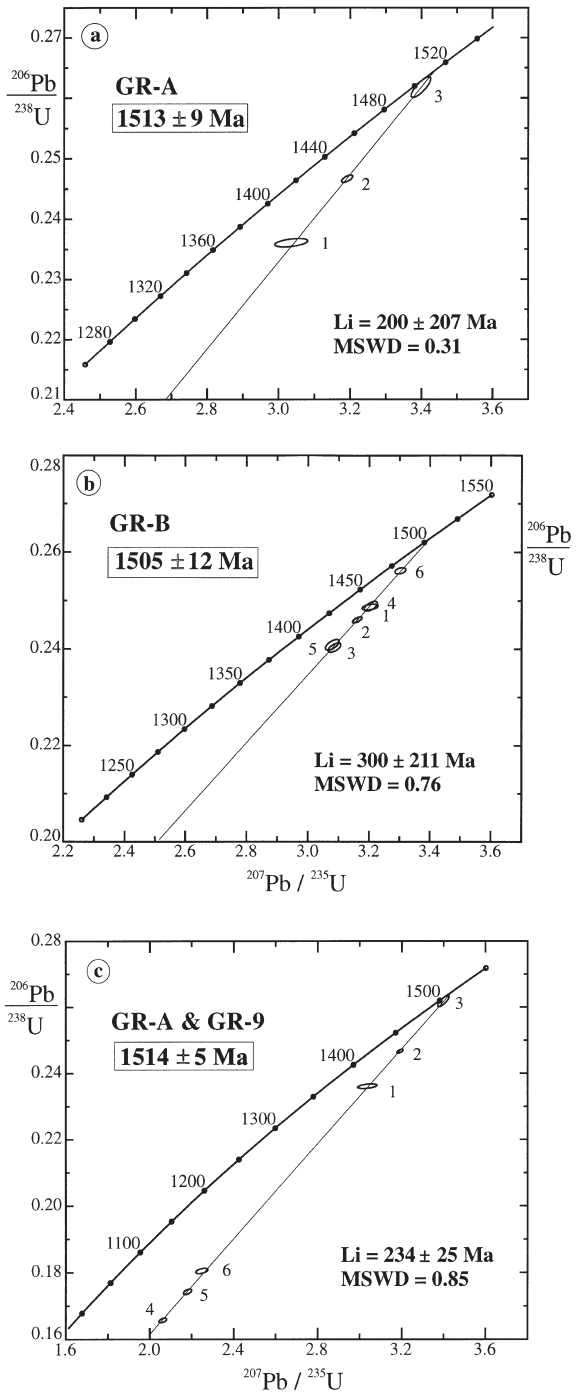


Fig. 7. Concordia diagrams showing data of the U–Pb analyses of zircon fractions from the granites of the Ragunda complex. (a) GR-A (amphibole–biotite granite, Western Massif); (b) GR-B (medium-grained, equigranular biotite granite, Central Massif); (c) GR-A and GR-9 (amphibole–biotite granite, Western Massif).

intrusion age of the amphibole–biotite granite. No significant change in upper intercept and error estimate results when omitting fractions 1 and/or 4 from the calculations.

5.2. Rb–Sr whole rock work

Kornfält (1976) reported a Rb–Sr age of 1320 ± 30 Ma for the silicic rocks of the Ragunda complex. Recalculation of this age using the decay constant recommended by Steiger and Jäger (1977) gives 1293 Ma. Kornfält (1976) sampled the entire outcrop area of the complex (i.e. syenites and granites, Kornfält, personal communication). For this work, a new attempt to date the rocks with the Rb–Sr whole-rock method was undertaken and sampling was restricted to two discrete intrusions, both in the Western Massif: a granite in the easternmost part of the massif (eight samples around the village of Ragunda) and a syenite on the western flank of the massif (three samples close to Lake Gesunden). An errorchron for the granite corresponds to an age of 1233 ± 91 Ma and is in the same range as reported by Kornfält (1976). This is 281 Ma lower than the zircon age from the same sample and evidently the Rb–Sr isotopic system of the granite has been disturbed after crystallization. On the contrary, the Rb–Sr age estimate of the syenite (1515 ± 71 Ma) is identical to the zircon age but is hampered by the large MSWD of 3.58. The disturbance of the Rb-Sr isotopic system of the granites may be due to the intrusion of the Postjotnian dolerite dykes at 1250–1200 Ma.

6. Discussion

6.1. Relative ages interpreted from the aeromagnetic map of the Ragunda complex

The total magnetization of a rock is mainly a function of its magnetite content (cf Thompson and Oldfield, 1986 and references therein). Thus, regional aeromagnetic surveys of magnetization intensity result in maps showing the regional variation in magnetite content. When regional magnetic

data are interpreted in terms of magnetic intensity, they can be used for tectonic analyses of the map area (e.g. Henkel, 1991). Structural analyses of aeromagnetic maps can give important information of the shape and age relation of various magnetic structures (truncation relations). In the Ragunda complex, the internal magnetic patterns, contacts and magnetization levels provide useful information about the internal structures and age relations between the different intrusions.

The Western Massif consists of a nearly linear sequence of approximately circular gabbro, syenite and granite intrusions, younging from west to east; the younger, more easterly intrusions cut through the older westerly intrusions. In the aeromagnetic map (Fig. 6), this can be seen as incomplete circles truncated in the east by younger circular intrusions. This interpretation suggests that the westernmost part of the massif is older than the amphibole–biotite granite at the village Ragunda. This amphibole–biotite granite gives a composite U–Pb upper intercept zircon age of 1514 ± 5 Ma, which, together with the magnetic pattern, indicates that the westernmost part of the massif should have intruded prior to 1509 Ma.

The magnetic pattern of the Central Massif does not show any obvious linear intrusion sequence but helps to resolve the lithologic differences between the Central and Western massifs. The highest magnetization levels are found mostly near the intrusion margins, which coincides with the presence of gabbro. Granite is more abundant in the Central Massif than in the Western Massif, whereas syenitic rocks are only found in the Western Massif. If there was a fault between the two massifs, different erosion levels could be expected. Kornfält (1976) reported that the rocks in the Central Massif are miarolitic to a higher degree than those in the Western Massif and include a granite porphyry. According to Kornfält (1976) this indicates that the Central Massif represents a more shallow section. During the mapping for the present contribution, however, a large number of miarolitic cavities were also found in the granites of the Western Massif. The westernmost amphibole-bearing granite has, for instance, large druses with well-developed fluorite and quartz crystals. This is also true for the biotite granite 2 km west of the village of Hammarstrand. No field evidence has been found to support the existence of a fault between the Western and Central Massifs, nor does the aeromagnetic map show any fault between them. The aeromagnetic pattern in the border zone between the Western and Central Massifs has a curved outline, which rather corresponds to separate circular intrusions.

In the Central Massif, early intrusions have probably been overprinted by later intrusions, in most cases of a more silicic composition. The latest intrusions can be found as intact circular bodies in the aeromagnetic map, although the age relation between them is unknown. In the northeastern part of the Central Massif there is a low magnetic area with a circular outline. This area consists of the biotite granite (GR-B) that was dated at 1505 ± 12 Ma. At face value, the age is lower than that of the amphibole–biotite granite (1514 ± 5 Ma) from the Western Massif; however, the two intrusions overlap in age within the error. No age difference can be suggested from the field evidence or the aeromagnetic map between the Western and Central Massifs either.

The Northeastern Massif stands out as a distinct magnetic anomaly immediately northeast of the Central Massif. Outcrops of metagreywacke are found between the Northeastern and Central Massifs; in the aeromagnetic map they show up as a low magnetization level between the massifs. The Northeastern Massif is a small centred complex with only two major intrusions, a gabbro intruded by a granite.

6.2. *Evidence for contemporaneity*

The rapakivi granites are often spatially and temporally associated with basic rocks (e.g. Rämö and Haapala, 1995). Evidence for coexisting and interacting silicic and basic magmas has recently been described from the Wiborg and Åland rapakivi areas (e.g. Rämö, 1991; Eklund, 1993; Salonsaari, 1995)

The Ragunda complex consists of three coeval but not comagmatic rock types (Persson, 1996):

(1) a gabbroic unit;
(2) a syenitic to peralkaline silicic unit; and

(3) a metaluminous to peraluminous silicic unit. The gabbro is generally occupying a marginal position. The reason for this may be that the relatively more low-viscous basic magma acted as a marginal lubricant in the magma conduits, and therefore was able to transport the more viscous intermediate and silicic magmas from their source to their final intrusion level (cf Blake and Campbell, 1986; Carrigan and Eichelberger, 1990; Carrigan, 1994). This would also explain the limited amount of mixing of the three magmas, as the more viscous magmas (silicic-intermediate) are relatively passively transported through the central parts of the conduit. Interaction may also have been restrained by differences in viscosity, temperature etc. (cf Sparks and Marshall, 1986; Frost and Mahood, 1987; Barbarin and Didier, 1992), and has only resulted in intermingling, which is quite widespread in the Ragunda complex.

Typical magma mingling structures have been observed in all the three massifs (Fig. 5). Field evidence shows that mingling must have been due to coexistence of granitic and gabbroic magmas. The gabbro shows chilled and crenulated margins against the granite. Some relationships, for example, apophyses of granite in gabbro, suggest that a fraction of the mafic rocks was partly crystallised when invaded by silicic magma. The fine-grained crenulate margins of the gabbroic rocks are regarded as the result of chilling of high-temperature, low-viscosity, mafic magma against the relatively cool and viscous silicic magma. Incomplete mixing of magmas has occurred on scales ranging from metre-sized pillows through centimetre-sized inclusions to complete dispersion with incorporation of xenocrysts into the silicic magma. The contrasted temperatures, viscosities, densities and various cooling rates of individual pillows are reflected in the diversity of textures commonly occurring in a single exposure [Fig. 5(b)]. Some of the dykes of the Ragunda complex show sharp and sinuous contacts to the rapakivi country rocks, indicating that the country rocks were only partly crystallised when intruded by the dykes. The composite silicic-mafic dykes also verify that both end members were at least partly liquid at the same time.

6.3. Regional significance of the Fennoscandian rapakivi complexes

The ages and the geographical distribution of the western Fennoscandian rapakivi complexes are shown in Fig. 2, except the Noran [1469 ± 10 Ma; Claesson and Kresten (1997)] and Strömsbro [1500 ± 19 Ma; Andersson (1997a)] intrusions that are found ca 300 km south of Ragunda and ca 100 km apart from each other. These complexes, excluding Nordingrå [1578 ± 17 Ma, Welin and Lundqvist (1984)], constitute the youngest (1.53–1.47 Ga) age group in the Fennoscandian Shield. Mullnäset in the northwest is the oldest of the dated complexes [1526 ± 3 Ma; Andersson and Neymark (1994)], while Noran is the youngest. The lowest age obtained from the Ragunda complex, 1505 ± 12 Ma, is basically the same as the age of the Rödön and Strömsbro complexes. The oldest age obtained from the Ragunda complex, 1514 ± 5 Ma, does not overlap with the ages obtained from the Mårdsjö complex (1524 ± 3 Ma) and the Mullnäset complex (1526 ± 3 Ma). Even though the oldest complexes are situated in the northwest (Fig. 2) and the Western Massif of the Ragunda complex shows a younging easterly trend, the age trend for the whole age group is not well defined. The three small complexes north of the Ragunda overlap in age, as do the Nordsjö complex and the Ragunda complex. The youngest age from the Ragunda complex overlaps, in turn, with the age of the Rödön and Strömsbro complexes. So even if there is a subtle indication of an age trend (younging from northwest to southeast), intrusions may have occurred contemporaneously in Ragunda, Rödön, Strömsbro and Nordsjö.

The rapakivi occurrences in the Fennoscandian Shield can be divided into three age groups, two of which show a pronounced linear distribution (Figs. 1 and 2). Excluding the Salmi complex, the ages become systematically younger from the east to the west. Within the age groups, no correlation between location and age exists. In the 1.59–1.56 Ga age group, the southernmost complex (Riga) has been dated to 1576 ± 3 and 1584 ± 7 Ma (Rämö et al., 1996); these ages overlap with the age of the Nordingrå complex (1578 ± 19 Ma). Rapakivi magmatism apparently

occurred over a relatively short interval of time across a crustal segment >700 km in length. Between the Riga and Nordingrå complexes, a complete overlap in ages is found. The Salmi complex in the easternmost part of the Fennoscandian Shield is younger than the 1.65–1.62 Ga Wiborg complex; reported U–Pb ages for the former are 1539±11 Ma (Suominen, 1991) and a 1543±8 and 1517±47 Ma (Neymark et al., 1994). The youngest age is close to the U–Pb ages of the Ragunda complex, suggesting that magmatic activity was roughly coeval at the eastern and western extremes. Thus, it is unlikely that the surface location of each granite centre describes a plate motion related to a fixed hot spot in the mantle. As stated by Haapala and Rämö (1992), this excludes continental plate overriding a single hot mantle plume as a possible explanation (cf Pesonen et al., 1989) for the genesis of rapakivi intrusions. Rapakivi anorthosite–granite complexes found in Precambrian shield areas generally occur along roughly linear intracontinental zones of considerable extent. Emplacement of these complexes is attributed by most authors to rifting events, incipient or aborted (e.g. Anderson, 1983; Emslie, 1985; Windley, 1983; Haapala and Rämö, 1992).

According to Haapala and Rämö (1990), the swarms of diabase (as well as minor quartz–feldspar porphyry) dykes temporally and spatially associated with the Finnish rapakivi granite complexes (see Laitakari and Leino, 1989; Vaasjoki et al., 1991; Ehlers and Ehlers, 1977; Suominen, 1991; Rämö, 1991) clearly demonstrate that the tectonic setting during their emplacement was extensional. The dyke swarms that are coeval with the older (1.65 to 1.62 Ga) Finnish rapakivi complexes in the east have an overall WNW-strike and roughly coincide with that of the Jotnian (1.3–1.4 Ga) sandstone-filled graben in southwestern Finland. In contrast, most of the dyke swarms associated with the younger (1.58 to 1.54 Ga) western Finnish rapakivi complexes strike nearly perpendicular to this trend. This shows that the extension direction varied with time or area (Rämö and Haapala, 1995).

The emplacement of the Ragunda complex and the small Swedish complexes probably also took place along deep fractures formed in an extensional regime. This is supported by the large number of diabase and quartz–feldspar porphyry dykes associated with the Ragunda complex; these generally trend NW–SE to E–W. Numerous dykes are also found between the Ragunda and Nordingrå complexes, as well as in connection with the Nordsjö complex, and generally show a NW–SE to E–W strike (Lundqvist et al., 1990). The dykes and the outline of the Western Massif in the Ragunda complex are approximately parallel to the major recognisable NW–SE and E–W lineaments in the Precambrian basement of central Sweden (cf Lundqvist et al., 1990). The lineament structure of the Ragunda area judged from the orientation of river valleys (Fig. 3; Indalsälven and Faxälven), also trend in a NW–SE and a E–W direction; the zigzag pattern from the Indalsälven constitutes a very pronounced lineament. The Ragunda complex and the smaller complexes in the western age group were probably intruded in a restricted time interval along the same crustal weakness trajectories.

7. Conclusions

The most important results of this field, magnetic and geochronologic study of the Ragunda rapakivi complex in central Sweden are:

(1) The Ragunda complex consists of three well-defined plutons — the Western Massif, the Central Massif and the Northeastern Massif.
(2) The Western Massif is a centred complex with a W–E polarity. Magmatic activity probably started before 1514 Ma in the west, continued along a linear intrusion sequence and ended in the east with the intrusion of the amphibole–biotite granite at 1514±5 Ma (U–Pb zircon age).
(3) The Central Massif is a large centred complex, with no obvious linear intrusion sequence. The Central Massif was emplaced as a large number of separate intrusions. This has resulted in a 'cloudy' aeromagnetic structure on the map. The earliest intrusions have been overprinted by later, in most cases more silicic,

intrusions. One late granite intrusion gave a U–Pb zircon age of 1505 ± 12 Ma.

(4) The Northeastern Massif is a small centred complex with two major intrusions, gabbro followed by granite.
(5) A significant age difference is not anticipated between the different intrusions in the Ragunda complex. Magma mingling structures are observed throughout the entire complex, most commonly in contact zones between gabbro and granite. Field evidence shows that the mingling must have been due to coexistence of granitic and gabbroic magmas.
(6) Together with the small complexes of Strömsund, Mullnäset, Nordsjö, Mårdsjö, Rödön, Strömsbro and Noran, the Ragunda complex forms the youngest group of Rapakivi complexes in the Fennoscandian Shield.
(7) The emplacement of the Ragunda complex and the small complexes in central Sweden cannot be explained by a plate moving over a stationary hot spot. The emplacement of the Ragunda complex and the small complexes probably occurred through deep crustal fractures formed in an extensional regime. This is supported by the large number of subparallel mafic and acid dykes found within and between the complexes in central Sweden.

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References

Anderson, J.L., Proterozoic anorogenic granite plutonism of North America. In: Medaris, L.G., Jr, Byers, C.W., Mickelson, D.M., Shanks, W.C. (Eds.), Proterozoic Geology: Selected Papers from an International Proterozoic Symposium. 1983. Geol. Soc. Am. Mem. 161, 133–154.

Andersson, U.B., 1997a. The sub-Jotnian Strömsbro granite complex at Gävle, Sweden. Geol. Fören. Stockholm Förh. 119, 159–167.

Andersson, U.B., 1997b. Petrogenesis of some Proterozoic granitoid suites and associated basic rocks in Sweden (geochemistry and isotope geology). Sveriges Geol. Unders., Rapporter och meddelanden 91, 216 pp.

Andersson, U.B., Neymark, L.A., Isotopic constraints on age and sources of some minor rapakivi complexes, central Sweden. In: Lanphere, M.A., Dalrymple, G.B., Turrin, B.D. (Eds.), Abstracts of the Eight International Conference on Geochronology, Cosmochronology, and Isotope Geology. 1994. U.S. Geol. Surv. Circ. 1107, 8

Barbarin, B., Didier, J., 1992. Genesis and evolution of mafic microgranular enclaves through various types of interaction between coexisting felsic and mafic magmas. Trans. R. Soc. Edinburgh: Earth Sci. 83, 145–153.

Blake, S., Campbell, I.H., 1986. The dynamics of magma-mixing during flow in volcanic conduits. Contrib. Miner. Petrol. 94, 72–81.

Carrigan, C.R., 1994. Two-component magma transport and the origin of composite intrusions and lava flows. In: Ryan, M.P. (Ed.), Magmatic Systems. Academic Press, New York, pp. 319–354.

Carrigan, C.R., Eichelberger, J.C., 1990. Zoning of magmas by viscosity in volcanic conduits. Nature 343, 248–254.

Claesson, S., Lundqvist, Th., 1995. Origins and ages of Proterozoic granitoids in the Bothnian Basin, central Sweden; isotopic and geochemical constraints. Lithos 36, 115–140.

Claesson, S., Kresten, P., 1997. The anorogenic Noran granite — a Mesoproterozoic rapakivi intrusion in south-central Sweden. Geol. Fören. Stockholm Förh. 119, 115–122.

Delin, H., 1993. The radiometric age of the Ljusdal granodiorite of central Sweden. Sveriges Geol. Unders., Ser. C 823, 13–16.

Ehlers, C., Ehlers, M., 1977. Shearing and multiple intrusion in the diabase of Åland archipelago, SW Finland. Geol. Surv. Finland Bull. 289, 31 pp.

Eklund, O., 1993. Coeval contrasting magmatism and magmatic mixing in Proterozoic post- and anorogenic granite, Åland, SW Finland. Department of Geology and Mineralogy, Åbo Akademi University, Åbo. Åbo Akademis tryckeri, Åbo, Finland, 57 pp.

Emslie, R.F., 1985. Proterozoic anorthosite massifs. In: Tobi, A.C., Touret, J.L.R. (Eds.), The Deep Proterozoic Crust in the North Atlantic Provinces. Reidel, Dordrecht, pp. 39–60.

Emslie, R.F., Hunt, P.A., 1990. Ages and petrogenetic significance of igneous mangerite–charnockite suites associated with massif anorthosites, Grenville province. J. Geol. 98, 213–231.

Erdmann, A., 1847. Gångar av granit, genomsättande hyperstenfels. Öfversigt af Kungl. Vet. Ak. förhandl (in Swedish).

Frost, T.P., Mahood, G.A., 1987. Field, chemical, and physical constraints on mafic–felsic magma interaction in the Lamarck Granodiotite, Sierra Nevada, California. Geol. Soc. Am. Bull. 99, 272–291.

Gorbatschev, R., Solyom, Z., Johansson, I., 1979. The Central Scandinavian Dolerite Group in Jämtland, central Sweden. Geol. Fören. Stockholm Förh. 101, 177–190.

Haapala, I., 1977. Petrography and geochemistry of the Eurajoki stock, a rapakivi–granite complex with greisen-type mineralization in southwestern Finland, Geol. Surv. Finland Bull. 286, 128 pp.

Haapala, I., Rämö, O.T., 1990. Petrogenesis of the Proterozoic rapakivi granites of Finland. In: Stein, H.J., Hannah, J.L. (Eds.), Ore-bearing Granite Systems; Petrogenesis and Mineralizing Processes (Spec. Paper). Geol. Soc. Am. 246, 275–286.

Haapala, I., Rämö, O.T., 1992. Tectonic setting and origin of the Proterozoic rapakivi granites of southeastern Fennoscandia. Trans. R. Soc. Edinburgh: Earth Sci. 83, 165–171.

Henkel, H., 1991. Magnetic crustal structures in northern Fennoscandia. Tectonophysics 192, 57–79.

Hill, J.D., Miller, R.R., 1990. A review of middle Proterozoic epigenic felsic magmatism in Labrador. In: Gower, C.F., Rivers, T., Ryan, A.B. (Eds.), Mid-Proterozoic Laurentia–Baltica (Spec. Paper). Geol. Assoc. Can. 38, 417–431.

Högbom, A.G., 1893. Om postarkäiska eruptiver inom det svensk-finska urberget. Geol. Fören. Stockholm Förh. 15, 209–240.(in Swedish)

Högbom, A.G., 1894. Geologisk beskrifning ofver Jämtlands län. Sveriges Geol. Unders. Ser. C 140, 106 pp.

Högbom, A.G., 1899. Om Ragundadalens geologi. Sveriges Geol. Unders. Ser. C 182, 121 pp.

Högbom, A.G., 1909. The igneous rocks of Ragunda, Alnö, Rödö och Nordingrå. Geol. Fören. Stockholm Förh. 31, 347–375.

Idman, H., 1989. The Siipyy granite — a new rapakivi occurrence in Finland. Bull. Geol. Soc. Finland 61, 123–127.

Kirs, J., Huhma, H., Haapala, I., 1991. Petrological–chemical features and age of Estonian anorogenic potassium granites. In: Haapala, I., Rämö, O.T. (Eds.), Symposium on Rapakivi Granites and Related Rocks. Abstract Volume. Geol. Surv. Finland Guide 34, 28–29.

Kornfält, K.-A., 1969. X-ray and optical observations on the K-feldspars from the Ragunda area, central Sweden. Sveriges Geol. Unders. Ser. C 636, 33 pp.

Kornfält, K.-A., 1976. Petrology of the Ragunda rapakivi massif, central Sweden. Sveriges Geol. Unders. Ser. C 725, 111 pp.

Krogh, T.E., 1973. A low-contamination method for hydrothermal decomposition of zircon and extraction of U and Pb for isotopic age determinations. Geochim. Cosmochim. Acta 37, 485–494.

Krogh, T.E., 1982. Improved accuracy of U–Pb zircon ages by the creation of more concordant systems using an abrasion technique. Geochim. Cosmochim. Acta 46, 637–649.

Laitakari, I., Leino, H., 1989. A new model for the emplacement of the Häme diabase dyke swarm, central Finland. In: Autio, S. (Ed.), Geological Survey of Finland, Current Research 1988 (Spec. Paper). Geol. Surv. Finland 10, 7–8.

Lindh, A., Johansson, I., 1996. Rapakivi granites of the Baltic Shield: the Nordingrå granite, its chemical variation and Sm–Nd isotope composition. N. Jahrb. Mineral. Abh. 170, 291–312.

Lundbohm, Hj., 1893. Om berggrunden i Vesternorrlands kusttrakter. Geol. Fören. Stockholm Förh. 15, 321–326.(in Swedish)

Lundbohm, Hj., 1899. Berggrunden inom Vesternorrlands län. Sveriges Geol. Unders. Ser. C 177, 60 pp. (in Swedish).

Lundegårdh, P.H., Gorbatschev, R., Kornfält, K.-A., 1984. Karta över berggrunden i Jämtlands län (urberget). Sveriges Geol. Unders. Ser. Ca 53 (mapsheet).

Lundqvist, Th., Gee, D.G., Kumpulainen, R., Karis, L., Kresten, P., 1990. Beskrivning till berggrundskartan över Västernorrlands län. With English Summary. Maps to the scale of 1:200 000. Sveriges Geol. Unders. Ser. Ba 31, 429 pp.

Ludwig, K.R., 1980. Calculation of uncertainties of U–Pb isotope data. Earth Planet. Sci. Lett. 46, 212–220.

Ludwig, K.R., 1991a. PBDAT. A computer program for processing Pb–U–Th isotope data. Version 1.20. U.S. Geol. Surv. Open-File Rep. 91.

Ludwig, K.R., 1991b. ISOPLOT. A plotting and regression program for radiogenic-isotope data. Version 2.53. U.S. Geol. Surv. Open-File Rep. 91.

Neymark, L.A., Amelin, J.V., Larin, A.M., 1994. Pb–Nd–Sr isotopic and geochemical constraints on the origin of the 1.54–1.56 Ga Salmi rapakivi granite–anorthosite batholith (Karelia Russia). Miner. Petrol. 50, 73–193.

Nurmi, P.A., Haapala, I., 1986. The Proterozoic granitoids of Finland: granite types, metallogeny and relation to crustal evolution. Bull. Geol. Soc. Finland 58, 203–233.

Persson, A.I., 1995. Geochemistry, petrology and geochronology of the Ragunda rapakivi complex, central Sweden. In:

Dall'Agnol, R., Macambria, M.J.B., Costi, H.T. (Eds.), Symposium on Rapakivi Granites and Related Rocks, Abstract Volume. Center for Geosciences, Federal University of Pará, Belem, Brazil, pp. 62–63.

Persson, A.I., 1996. Petrogenesis of the Ragunda rapakivi complex, central Sweden. In: Haapala, I., Rämö, O.T., Kosunen, P. (Eds.), Symposium on Rapakivi Granites and Related Rocks, Abstract Volume. University of Helsinki. University Press, Helsinki, Finland, p. 57.

Persson, L., 1978. The Revsund–Sörvik granites in the western parts of the province of Ångermanland, central Sweden. Sveriges Geol. Unders. Ser. C 741, 1–59.

Persson, P.-O., Wahlgren, C.-H., Hansen, B.T., 1983. U–Pb ages of Proterozoic metaplutonics in the gneiss complex of southern Värmland, south-western Sweden. Geol. Fören. Stockholm Förh. 105, 1–8.

Pesonen, L.J., Torsvik, T.H., Elming, S.-Å., Bylund, G., 1989. Crustal evolution of Fennoscandia-palaeomagnetic constraints. Tectonophysics 162, 27–49.

Rämö, O.T., 1991. Petrogenesis of the Proterozoic rapakivi granites and related basic rocks of southeastern Fennoscandia: Nd and Pb isotopic and general geochemical constraints. Geol. Surv. Finland Bull. 355, 161 pp.

Rämö, O.T., Haapala, I., 1995. One hundred years of rapakivi granite. Miner. Petrol. 52, 129–185.

Rämö, O.T., Huhma, H., Kirs, J., 1996. Radiogenic isotopes of the Estonian and Latvian rapakivi granite suites: new data from the concealed Precambrian of the East European Craton. Precambrian Res. 79, 209–226.

Sahama, Th.G., 1945. On the chemistry of the east Fennoscandian rapakivi granites. Bull. Commun. geol. Finlande 136, 15–67.

Salonsaari, P.T., 1995. Hybridization in the subvolcanic Jaala–Iitti complex and its petrogenetic relation to rapakivi granites and associated mafic rocks of southeastern Finland. Bull. Geol. Soc. Finland 67 Part 1b.

Sederholm, J.J., 1891. Ueber die finnlädischen Rapakiwigesteine. Tschermak's Mineral. Petrograph. Mitth. 12, 1–31.

Sobral, J.M., 1913. Contributions to the geology of the Nordingrå region. Ph.D. thesis, Uppsala University, Sweden.

Sparks, R.S.J., Marshall, L.A., 1986. Thermal and mechanical constraints of mixing between mafic and silicic magmas. J. Volc. Geotherm. Res. 29, 99–124.

Stacey, J.S., Kramers, J.D., 1975. Approximation of terrestrial lead isotope evolution by a two-stage model. Earth Planet. Sci. Lett. 26, 207–221.

Steiger, R.H., Jäger, E., 1977. Subcommission on Geochronology: convention on use decay constants in geo- and cosmochronology. Earth Planet. Sci. Lett. 36, 359–362.

Suominen, V., 1991. The chronostratigraphy of southwestern Finland with special reference to Postjotnian and Subjotnian diabases. Geol. Surv. Finland Bull. 356, 100 pp.

Thompson, R.N., Oldfield, F., 1986. Environmental Magnetism. Allen and Unwin, London, Winchester.

Todt, W., Cliff, R.A., Hanser, A., Hofmann, A.W., 1993. Recalibration of NBS lead standards using a $^{202}Pb+^{205}Pb$ double spike. Terra Nova 5 (EUG7 Abstracts, Abstract Supplement No. 1), 396

Vaasjoki, M., 1977. Rapakivi granites and other postorogenic rocks in Finland: Their age and the lead isotopic composition of certain associated galena mineralizations. Geol. Surv. Finland Bull. 294, 64 pp.

Vaasjoki, M., Pihlaja, P., Sakko, M., 1988. The radiometric age of the Reposaari granite and its bearing on the extent of the Laitila rapakivi batholith in western Finland. Bull. Geol. Soc. Finland 60, 129–134.

Vaasjoki, M., Rämö, O.T., Sakko, M., 1991. New U–Pb ages from the Wiborg rapakivi area: constraints on the temporal evolution of the rapakivi granite–anorthosite–diabase dyke association of southeastern Finland. Precambrian Res. 51, 227–243.

Vaasjoki, M., Rämö, O.T., Alviola, R., Johanson, B.S., 1993. Petrography and new U–Pb age data on the Ahvenisto rapakivi complex, southeastern Finland. Geol. Soc. Am. Abstr. Progr. 25, 86

Vorma, A., 1976. On the petrochemistry of rapakivi granites with special reference to the Laitila massif, southwestern Finland. Geol. Surv. Finland Bull. 285, 98 pp.

Wahl, W., 1925. Die Gesteine des wiborger Rapakiwigebietes. Fennia 45 (20), 127 pp.

Wasström, A., 1993. The Knaften granitoids of Västerbotten County, northern Sweden. Sveriges Geol. Unders. Ser. C 823, 60–64.

Welin, E., 1979. Tabulation of recalculated radiometric ages of extrusive and intrusive rocks in northern Sweden. Geol. Fören. Stockholm Förh. 101, 309–320.

Welin, E., 1994. The U–Pb age of the Rödön rapakivi granite, central Sweden. Geol. Fören. Stockholm Förh. 116, 113–114.

Welin, E., Lundqvist, Th., 1975. K–Ar ages of Jotnian dolerites in the Västernorrland County, Central Sweden. Geol. Fören. Stockholm Förh. 97, 83–88.

Welin, E., Lundqvist, Th., 1984. Isotopic investigation of the Nordingrå rapakivi massif, north-central Sweden. Geol. Fören. Stockholm Förh. 106, 41–49.

Welin, E., Christiansson, K., Kähr, A.-M., 1993. Isotopic investigations of metasedimentary and igneous rocks in the Paleoproterozoic Bothnian basin, central Sweden. Geol. Fören. Stockholm Förh. 115, 285–296.

Windley, B.F., 1983. A tectonic review of the Proterozoic. In: Medaris, L.G., Jr, Byers, C.W., Mickelson, D.M., Shanks, W.C. (Eds.), Proterozoic Geology: Selected Papers from an International Proterozoic Symposium. Geol. Soc. Am. Mem. 161, 1–10.